Chronos T5 Tiny vs. Classical Forecasting Methods

An applied foundation-model comparison across six datasets, nine models, and five metrics

Foundation: Chronos T5 Tiny

Classical: seven methods

Neural reference: DeepAR

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What the project compares

Evaluate whether Chronos T5 Tiny, a pretrained **foundation time-series model**, can match or outperform **classical forecasting methods** under one common pipeline. DeepAR remains a separate neural reference.

Accuracy comparison

Which model minimizes median-forecast error?

Uncertainty comparison

Which model produces the best quantile forecasts?

Practical tradeoffs

Is the comparison valid, and what does each method cost to run?

Six datasets, nine models, one common pipeline

6 datasets two source families → **71 tasks** one final holdout each × **9 models** same tasks and metrics = **639 results** zero failed rows

Dataset	Tasks	Horizon	Lag
M1 yearly	20	6	1
M4 hourly	20	48	24
Weather	20	30	7
National illness	1	24	52
Traffic	5	24	24
PSM	5	24	24

Model families

- **Foundation model**: Chronos T5 Tiny
- **Classical methods**: seasonal naive, AR, MA, ARMA, ARIMA, AutoARIMA, Prophet
- **Additional neural reference**: DeepAR

CPU execution; point metrics use the median forecast.

Read the metric before reading the ranking

Point forecast metrics

MAE	Original units; easy to explain, scale-dependent.
RMSE	Original units; emphasizes large misses.
SMAPE	Range 0–2 here; unstable around zero.
MASE	Relative to an in-sample seasonal-naive scale.

MASE < 1: below the in-sample naive error scale

Probabilistic metric

WQL averages pinball loss across quantiles $0.1, \ldots, 0.9$ and normalizes by total absolute target.

- Dimensionless and unbounded above.
- Official aggregation weights by target magnitude.
- Chronos/DeepAR use samples; ARIMA uses Gaussian intervals.
- Seasonal naive repeats one point at every quantile.

All five metrics are lower-is-better, but they answer different questions.

There is no single winner across all forecasting tasks

Dataset	Point result	WQL winner	Main caveat
M1 yearly	DeepAR raw error; ARIMA scaled error	**DeepAR**	Largest series dominates
M4 hourly	Seasonal naive raw; Chronos scaled	**Chronos**	Objective changes winner
Weather	**No clean winner**	Chronos, narrowly	Chronos point forecast is zero-like
National illness	**ARIMA(2,1,2)**	**Prophet**	One series, one cutoff
PSM	**ARIMA(2,1,2)**	**Chronos**	Minute data labeled hourly
Traffic	**DeepAR**	**DeepAR**	First five sensors only

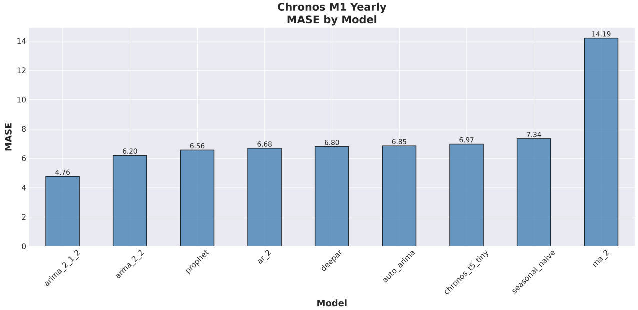
Best classical reference

ARIMA(2,1,2), mean rank 2.86 after excluding invalid weather SMAPE.

Foundation-model result

Chronos is strong on M4 and selected WQL comparisons, but not everywhere.

M1 yearly: winner depends on how series are weighted



Raw-scale winner

DeepAR: MAE **445,885**, RMSE **543,050**, official WQL **0.193**.

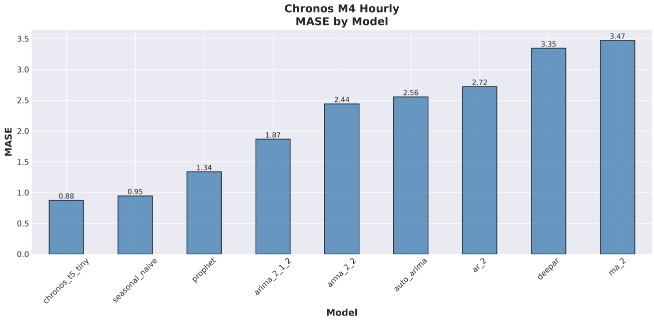
Scaled-error winner

ARIMA(2,1,2): SMAPE **0.216**, MASE **4.764**.

Interpret carefully

- Largest series carries 76.1% of WQL target weight.
- Every model has average MASE above 1.
- MA(2) WQL 131.1 is a variance explosion.

M4 hourly: Chronos and seasonal naive separate the field



Seasonal naive

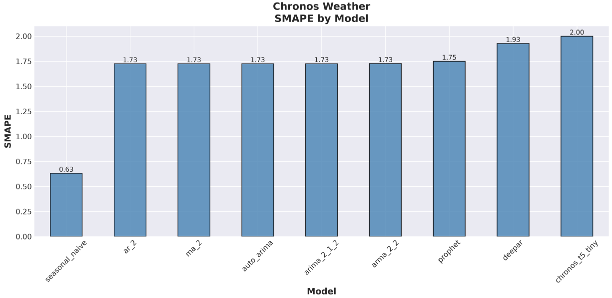
Best MAE **473.3** and RMSE **590.7**; total runtime about **0.01 s**.

Chronos

Best SMAPE **0.0452**, MASE **0.875**, and official WQL **0.0288**.

- Chronos wins WQL on 15 of 20 series.
- Both leaders have MASE below 1.
- Prophet is a distant but consistent third.

Weather: the apparent winner predicts almost exactly zero



Metric pathology

Chronos SMAPE is exactly **2.0** for all 20 series. Reproduced median predictions are only 1.3×10^{-8} to 4.6×10^{-8}.

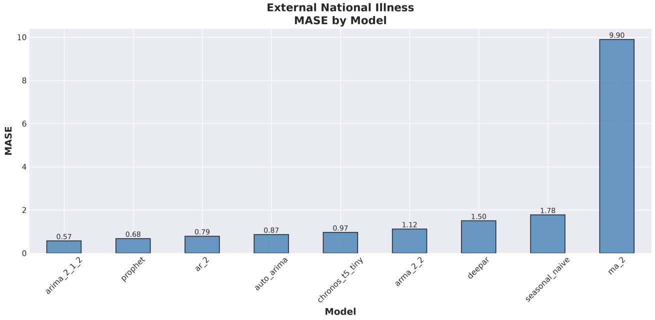
Why MAE still looks good

Rainfall is 72.67% zero. An exact-zero forecast matches Chronos MAE, RMSE, and MASE to floating-point precision.

- Chronos WQL 0.984 vs. zero baseline 1.000.
- DeepAR has the best RMSE, 1.965.
- Add rain-occurrence and positive-amount metrics.

Conclusion: no defensible overall weather winner yet.

National illness: classical models lead one test case



Point forecasts

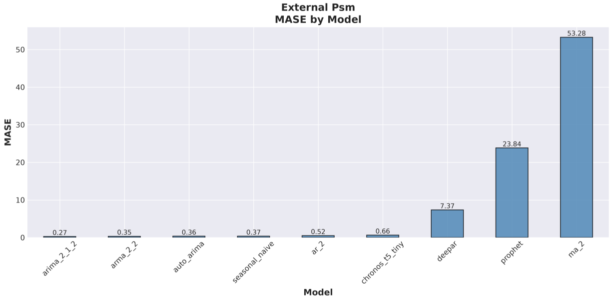
ARIMA(2,1,2) wins MAE, RMSE, SMAPE, and MASE. Its MASE is **0.571**.

Quantile forecasts

Prophet WQL is **0.0304**, about 9.1% below ARIMA's **0.0334**.

- Chronos ranks fifth by MASE: 0.968.
- MA(2) is last by a wide margin.
- Only one `OT` series and one cutoff were tested.

PSM: strong numbers, wrong operational time scale



Numerical ranking

ARIMA(2,1,2) wins all point metrics: MASE **0.270**. Chronos wins official WQL: **0.00375**.

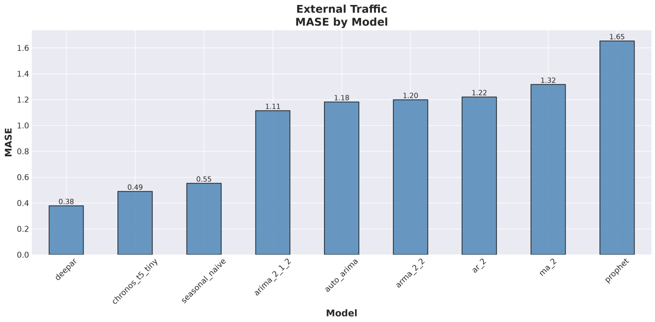
Configuration mismatch

Source `timestamp_(min)` increments by one minute, but the project code creates hourly timestamps and uses lag 24.

- This is a 24-step result, apparently 24 minutes.
- Large MASE values reflect tiny lag-24 scales.
- Five of 25 features were evaluated.

Do not select a PSM model until the time scale is corrected and rerun.

Traffic: DeepAR wins; Chronos remains competitive



DeepAR

Wins MAE **0.00765**, RMSE **0.01057**, MASE **0.378**, and WQL **0.1466**.

Alternatives

Chronos wins SMAPE at **0.198**; seasonal naive has MASE **0.552** at essentially zero runtime.

- DeepAR wins four of five sensors.
- Top three models all have MASE below 1.
- Prophet is last on every metric.

Scope: the first five columns out of 862 traffic signals.

Descriptive scorecard: ARIMA ranks first, Chronos second

Model	Mean rank*	Observed strength	Observed weakness
ARIMA(2,1,2)	**2.86**	Illness/PSM point; M1 scaled	M4, weather, traffic
Chronos T5 Tiny	**3.52**	M4; PSM WQL; traffic	M1; zero-like weather median
DeepAR	**4.62**	M1 weighted; traffic	M4 and PSM
Prophet	**4.69**	M1; illness WQL	PSM and traffic
Seasonal naive	**4.72**	Speed; M4; strong baseline	No calibrated uncertainty
AutoARIMA	**4.79**	Illness/PSM point	No metric wins; costly search
ARMA(2,2)	**4.90**	M1/PSM scaled error	Slow; no metric wins
AR(2)	**6.07**	Simple illness baseline	Mostly lower-middle
MA(2)	**8.83**	None demonstrated	Weak and unstable uncertainty

*Mean within-dataset rank over all metric cells except invalid weather SMAPE; descriptive project summary, with correlated metrics and unequal sample sizes.

What the comparison supports, and what it does not

What the comparison shows

- No model dominates every dataset and objective.
- ARIMA(2,1,2) is the strongest classical reference.
- Chronos is strong on M4 and probabilistic PSM scoring.
- DeepAR is strongest on the tested traffic sensors.
- Seasonal naive remains difficult to beat per unit runtime.

Project constraints

- One final holdout window per series.
- Only one illness target and five external columns.
- No repeated stochastic seeds or confidence intervals.
- M1 scale concentration changes the winner.
- Weather and PSM require evaluation fixes.

Use these rankings for the configured datasets and cutoffs only.

How to choose a model, and what to fix next

Speed / baseline	Point accuracy	Probabilistic quality	Weather or PSM?
Seasonal naive	ARIMA(2,1,2)	Compare WQL by dataset	Repair evaluation first

Repair validity

- Correct PSM timestamps and seasonal lag.
- Add zero and occurrence baselines for rainfall.
- Align WQL plotting with official aggregation.

Expand the comparison

- Use rolling origins and multiple seeds.
- Sample sensors/features rather than first columns.
- Persist forecasts and test interval calibration.

Chronos expands the toolkit; classical methods remain essential.

Use validated metrics to choose by dataset,
and keep strong classical baselines in every comparison.

Metric-aware | Baseline-grounded | Validity-checked

Amirhossein Bagheri | **PMDS Forecasting Project**

Thank you